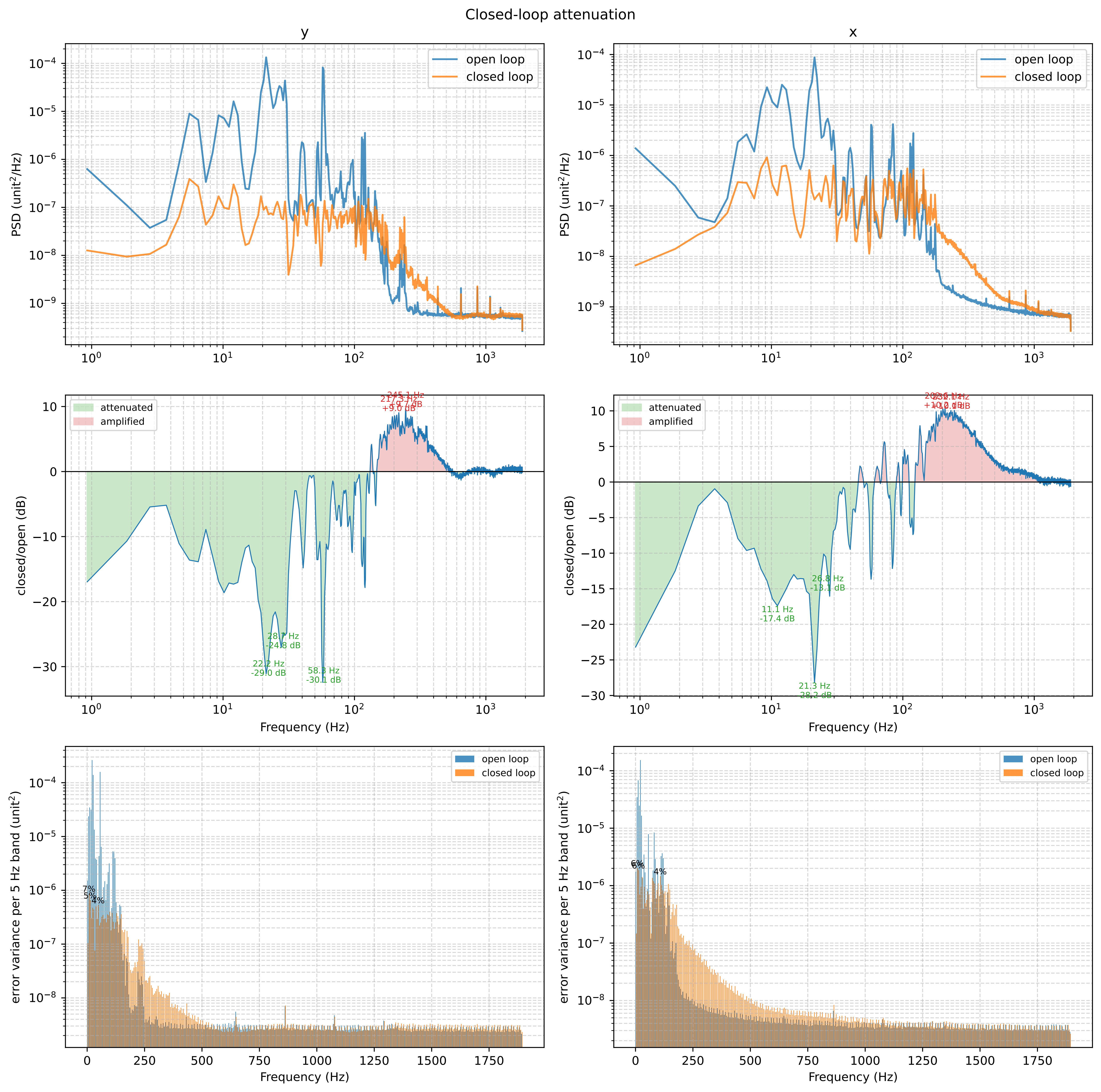
Closed-loop attenuation
y
x
open loop
closed loop
PSD (unit^2/Hz)
attenuated
amplified
closed/open (dB)
Frequency (Hz)
245.1 Hz
+9.0 dB
28.7 Hz
-24.8 dB
22.2 Hz
-29.0 dB
58.8 Hz
-30.1 dB
26.8 Hz
-13.1 dB
11.1 Hz
-17.4 dB
21.3 Hz
-28.2 dB
error variance per 5 Hz band (unit^2)
7%
5%
4%
6%

Run configuration and per-band RMS

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Started: 2026-07-30 16:13:25
pixel_scale: 15.5 px per output unit
y: mean -1.743 -> +0.000 px;  RMS about mean 0.469 -> 0.056 px   (open -> closed)
x: mean +3.534 -> -0.000 px;  RMS about mean 0.316 -> 0.089 px   (open -> closed)

Config used:
ATTENUATION PAR-3 -- BASELINE REPEAT of PAR-1 (identical fsp settings), the A in an A/B/A through the PARPORT DAC (TI DAC81404 over the StarTech PEX1P2 / AX99100 PCIe parallel card, spi link, mmio backend), replacing the DAQC2/BRIDGEplate serial path used by every previous attenuation run.
TRANSPORT: bit-banged 24-bit SPI, ~6.4 us per channel update (~12.8 us for both channels), against the BRIDGEplate's ~0.5-1.4 ms. DAC range is FIXED at +-5 V bipolar (153 uV/LSB) with the mirror parked at 0 V, matching the bode sweeps exactly; auto-ranging is off, so the actuator is a fixed-gain element for both phases of the A/B and the open/closed comparison is clean. Supplies +/-12 V: AVDD needs VMAX+1.5 = 11.5 V for +-10, AVSS -11.5 V, leaving only 0.5 V of margin -- meter the rails under load. AVSS is therefore NEGATIVE, not grounded, which means the 0-5 base range runs outside the AVSS = 0 V condition SLASEH2A 7.5 gives the unipolar figures at (see _avss_tradeoff); 16-bit = 76 uV/LSB in 0-5, 305 uV/LSB in +-10, vs the BRIDGEplate's 12-bit 1 mV.
OUTPUTS: two channels, no coarse stage -- ch0 = y/tip (+1.0 V/unit), ch1 = x/tilt (-1.0 V/unit), offset 0 V, symmetric command bounds +-1 V (59 px of authority against a +-16 px ROI; the mirror's full range is deliberately NOT given to the loop on this first run). NOTE the command->FSM voltage mapping is NOT the same as any previous attenuation run, so K is not comparable across the transport change and was re-measured -- and it came out 4.7-5.5x HIGHER, not lower as this file originally predicted, so nothing about the piplate-era gains carries over.
Camera/CoM: 32x32 ROI start_x 1840 / start_y 680, exposure 50, gain 100, bw 100, 20x20 tracked window thr 5 (the camera is now shrouded, so the background floor no longer needs a high threshold).
FSP: broad_order 512, broad_mu .03 -- REVERTED from PAR-2's 128/.005, which was 60% worse on y and 10% on x once normalised for the environment. 128 taps = 29.6 Hz resolution cannot separate the 2-3 Hz-spaced 9-30 Hz comb; this reproduces the documented kalman order-80 failure (order 80 ~29 Hz could not represent the 2.256 Hz comb). broad_lp 5 (retuned: the shorter delay moved the regeneration band from ~325 Hz to ~700 Hz); adapt_period 8, adapt_tau 5. Guard ON (guard_ratio 4). Cold-start observer: pass_open false + broad_freeze_closed false, so no weight is derived from open-loop data; drift_tau and the transient path are OFF. Modal freqs re-derived from run 1's own open PSD (8-mode cap), though they are inert while broad_order > 0.
PURPOSE: PAR-1 and PAR-2 were an hour apart and the environment moved between them (y open 5-10 Hz rose 4x), so this repeats PAR-1's exact controller to confirm the baseline and bound the run-to-run spread before any further tuning. PAR-1: y 0.518 -> 0.0804 px, x 0.496 -> 0.0966 px. PAR-2 (128/.005), normalised to PAR-1's environment: y 0.128, x 0.106; 60-150 Hz only 1.5x and the largest closed contributor on both axes; waterbed 3.4-3.5x at 195-250 Hz; no regeneration (400-800 Hz = 0.0% of command power).
DELAY AND K: BOTH re-fit for this transport 2026-07-30 from conf_bode_par_x/y.json (+/-100 mV sweep, full-band 5-300 Hz phase slope for tau, <=30 Hz plateau for K). Kx 3.787 +/- 0.008, Ky 3.636 +/- 0.005; delay x 2.717 fr, y 2.577 fr at fs 3788. Against the piplate era that is 4.7-5.5x MORE loop gain and LESS THAN HALF the delay -- the shorter delay is the entire reason to expect better rejection here.
COMPARE against attenuation_fsp21 and attenuation_fsp15 (run 15: y 0.073 / x 0.168 px): report total and per-band closed RMS, 60-400 Hz waterbed, command/clamp behavior, and transient activation count on the shared 500/500/4096 grid. Closed RMS in PIXELS remains comparable across the change; command units do not. refs: Kulcsar 2006 / Petit 2008 / Meimon 2010 / Correia 2010
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RMS contributed by each 10 Hz band (px)

band (Hz)	y open	y closed	x open	x closed
0-10	0.0778	0.0153	0.0932	0.0229
10-20	0.1261	0.0152	0.1493	0.0248
20-30	0.3105	0.0148	0.2018	0.0238
30-40	0.0644	0.0118	0.0341	0.0180
40-50	0.0323	0.0141	0.0226	0.0159
50-60	0.1984	0.0106	0.0459	0.0190
60-70	0.0412	0.0127	0.0108	0.0117
70-80	0.0251	0.0119	0.0237	0.0248
80-90	0.0217	0.0121	0.0521	0.0202
90-100	0.0357	0.0134	0.0234	0.0216
100-110	0.0138	0.0116	0.0156	0.0226
110-120	0.0502	0.0104	0.0408	0.0211
120-130	0.0331	0.0123	0.0288	0.0191
130-140	0.0101	0.0116	0.0090	0.0168
140-150	0.0158	0.0126	0.0170	0.0215
150-160	0.0060	0.0091	0.0059	0.0121
160-170	0.0045	0.0077	0.0062	0.0137
170-180	0.0049	0.0086	0.0063	0.0143
180-190	0.0021	0.0044	0.0035	0.0095
190-200	0.0016	0.0038	0.0028	0.0082
200-210	0.0018	0.0044	0.0025	0.0079
210-220	0.0018	0.0045	0.0023	0.0068
220-230	0.0036	0.0072	0.0024	0.0069
230-240	0.0032	0.0069	0.0023	0.0068
240-250	0.0024	0.0056	0.0022	0.0064
250-260	0.0013	0.0032	0.0021	0.0061
260-270	0.0013	0.0032	0.0020	0.0054
270-280	0.0014	0.0032	0.0020	0.0054
280-290	0.0013	0.0025	0.0020	0.0051
290-300	0.0013	0.0025	0.0019	0.0049
total 0-300	0.4138	0.0540	0.2876	0.0861